

Problem 1 (Linear Regression) Let's implement a tool needed for a computer vision algorithm called photometric stereo, which allows us to reconstruct a 3D object's surface from 2D images of it under different lighting conditions. Figure 1 gives a sneak peek at what you'll be able to do after completing all the pieces. We want to measure the 3D surface reconstruction shown in Figure 1 right. We model the object's surface as $z = f(x, y) \in \mathbb{R}_+$ where $x, y \in \mathbb{R}$ denotes coordinates on the table, and z is the height of the surface above the table. To measure the object, we change the light source orientation $\mathbf{l} \in \mathbb{R}^{3 \times 1}$ (a unit-norm vector). For each orientation $\mathbf{l}$, we could obtain a pixel intensity image $I(x, y)$. Under a diffuse reflection assumption, the following relationship holds:

$$I(x, y) = \alpha(x, y) \mathbf{l}^T \mathbf{n}(x, y), \quad (1)$$

where $\alpha(x, y) \in \mathbb{R}$ is related to the surface's own property and $\mathbf{n}(x, y) \in \mathbb{R}^3$ is the unit-norm surface normal vector of $f(x, y)$. Assume that for each coordinate (x, y) in $f(x, y)$, it is corresponding to a pixel (x, y) in an $I(x, y)$. Also assume the light is directional light, i.e., the orientation $\mathbf{l}$ for the light is the same for all pixels (x, y) .

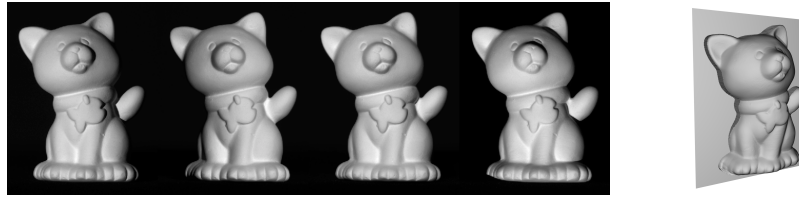


Figure 1: Photometric stereo. Left: 2D-pixel intensity $I(x, y)$ of a common scene under different lighting conditions. Right: 3D surface reconstruction $f(x, y)$ computed from the input images.

- (a) Suppose moving the light source in $\mathbf{l}_1, \dots, \mathbf{l}_d$, d different directions and obtain $\mathbf{I}_1, \dots, \mathbf{I}_d \in \mathbb{R}_+^{M \times N}$ 2D-pixel intensities matrix. We want to find the surface normal direction matrix $\mathbf{N} \in \mathbb{R}^{3 \times M \times N}$. Formulate this problem as a least squares problem.

Solution: Let us first reshape the matrix of $\mathbf{I}$ in to a vector, $\mathbf{I} \in \mathbb{R}_+^{1 \times MN}$, and correspondingly reshape the surface normal direction matrix $\mathbf{N}$ into $\mathbf{N} \in \mathbb{R}^{3 \times MN}$. So we could rewrite the Equation (1) into a matrix form:

$$\mathbf{I}_i = \mathbf{l}_i^T \underbrace{(\boldsymbol{\alpha} \odot \mathbf{N})}_{=: \boldsymbol{\rho} \in \mathbb{R}^{3 \times MN}}, i \in [d]$$

where we arrange $\alpha(x, y)$ into matrix $\boldsymbol{\alpha} \in \mathbb{R}^{3 \times MN}$ and $\odot$ denotes the element-wise multiplication. Furthermore, let us concatenate results from $\mathbf{l}_1, \dots, \mathbf{l}_d$ together, we get:

$$\underbrace{\begin{bmatrix} \mathbf{I}_1 \\ \vdots \\ \mathbf{I}_d \end{bmatrix}}_{=: \mathbf{I} \in \mathbb{R}^{d \times MN}} \approx \underbrace{\begin{bmatrix} \mathbf{l}_1 & \dots & \mathbf{l}_d \end{bmatrix}^T}_{=: \mathbf{L}^T, \mathbf{L} \in \mathbb{R}^{3 \times d}} \boldsymbol{\rho}.$$

Therefore, the least squares problem could be formulated as:

$$\hat{\boldsymbol{\rho}} = \arg \min_{\boldsymbol{\rho}} \|\mathbf{I} - \mathbf{L}^T \boldsymbol{\rho}\|_F^2.$$

And the surface norm $\mathbf{n}(x, y)$ could be calculated as:

$$\mathbf{n}(x, y) = \frac{\boldsymbol{\rho}(x, y)}{\|\boldsymbol{\rho}(x, y)\|_2}$$

Advance discussion: Here both $\mathbf{I}$ and $\mathbf{L}^T \boldsymbol{\rho}$ are matrix instead of vector, they are different from what we are learning in the course, so here we are using frobenius norm instead of L-2 norm. From the definition, given a matrix $\mathbf{A}$ frobenius norm of a matrix is defined as:

$$\|\mathbf{A}\|_F = \sqrt{\sum_{ij} A_{ij}^2} = \sqrt{\text{tr}(\mathbf{A}\mathbf{A}^T)}$$

So is the solution of $\arg \min_{\boldsymbol{\rho}} \|\mathbf{I} - \mathbf{L}^T \boldsymbol{\rho}\|_F^2$ equivalent to what we learned in the course? i.e. $\hat{\boldsymbol{\rho}} = (\mathbf{L}^T)^\dagger \mathbf{I}$. The answer is yes, but the derivation is out of the scope of this course. Here is the proof. First, we calculate the gradient of $\|\mathbf{I} - \mathbf{L}^T \boldsymbol{\rho}\|_F^2$ w.r.t $\boldsymbol{\rho}$.

$$\begin{aligned} \nabla_{\boldsymbol{\rho}} \|\mathbf{I} - \mathbf{L}^T \boldsymbol{\rho}\|_F^2 &= \nabla_{\boldsymbol{\rho}} \text{tr} \left((\mathbf{I} - \mathbf{L}^T \boldsymbol{\rho}) (\mathbf{I} - \mathbf{L}^T \boldsymbol{\rho})^T \right) \\ &= 2\mathbf{L} (\mathbf{L}^T \boldsymbol{\rho} - \mathbf{I}) \end{aligned}$$

How do I get the gradient? We could definitely derive it. But sometimes the method introduced in the first discussion won't work. To make life easier, you will find this [page](#) useful. Also, I personally use this [page](#) to help me calculate some uncommon matrix calculus. And then, set the gradient to zero, we get:

$$\mathbf{L} (\mathbf{L}^T \hat{\boldsymbol{\rho}} - \mathbf{I}) = 0 \Rightarrow \hat{\boldsymbol{\rho}} = (\mathbf{L}\mathbf{L}^T)^{-1} \mathbf{L}\mathbf{I} = (\mathbf{L}^T)^\dagger \mathbf{I}$$

- (b) Recovering the surface matrix $\mathbf{z} \in \mathbb{R}_+^{M \times N}$ from the surface normal direction matrix $\mathbf{N} \in \mathbb{R}^{3 \times M \times N}$. Formulate this problem as a least squares problem.

Hint 1: The relationship between surface norm $\mathbf{n}(x, y)$ and the object surface function $f(x, y)$ is:

$$\mathbf{n}(x, y) = \frac{\left[-\frac{\partial f(x, y)}{\partial x}, -\frac{\partial f(x, y)}{\partial y}, 1 \right]^T}{\sqrt{1 + \left(\frac{\partial f(x, y)}{\partial x} \right)^2 + \left(\frac{\partial f(x, y)}{\partial y} \right)^2}}.$$

Hint 2: The $f(x, y)$ could also be viewed as an image, such that:

$$\begin{aligned} \frac{\partial f(x, y)}{\partial x} &\approx \frac{f(x+1, y) - f(x, y)}{(x+1) - x} = f(x+1, y) - f(x, y) \\ \frac{\partial f(x, y)}{\partial y} &\approx \frac{f(x, y+1) - f(x, y)}{(y+1) - y} = f(x, y+1) - f(x, y) \end{aligned}$$

Solution: From this relationship, we could obtain:

$$\begin{aligned} \frac{\partial f(x, y)}{\partial x} &= -\frac{n_1(x, y)}{n_3(x, y)} \\ \frac{\partial f(x, y)}{\partial y} &= -\frac{n_2(x, y)}{n_3(x, y)} \end{aligned}$$

Let's define $\mathbf{N}_x \in \mathbb{R}^{M \times N}$ such that $\mathbf{N}_x[i, j] = -\frac{\mathbf{N}[0, i, j]}{\mathbf{N}[2, i, j]}$. So $\mathbf{N}_x$ represents the partial gradient of $\mathbf{z}$ related to x . Similarly, we define $\mathbf{N}_y$ represents the partial gradient of $\mathbf{z}$ related to y . This indicates that $\mathbf{N}_x[x, y] = \mathbf{z}[x+1, y] - \mathbf{z}[x, y]$ and $\mathbf{N}_y[x, y] = \mathbf{z}[x, y+1] - \mathbf{z}[x, y]$. As a result, we could formulate the problem as:

$$\hat{\mathbf{z}} = \arg \min_{\mathbf{z}} \left\| \begin{bmatrix} \mathbf{N}_x \\ \mathbf{N}_y \end{bmatrix} - \mathbf{D}\mathbf{z} \right\|_2^2$$

Here we vectorize the $\mathbf{N}_x$, $\mathbf{N}_y$ and $\mathbf{z}$, such that $\begin{bmatrix} \mathbf{N}_x \\ \mathbf{N}_y \end{bmatrix} \in \mathbb{R}^{2MN \times 1}$, $\mathbf{z} \in \mathbb{R}^{MN \times 1}$. Thus $\mathbf{D} \in \mathbb{R}^{2MN \times MN}$ and follows $\mathbf{N}_x[x, y] = \mathbf{z}[x+1, y] - \mathbf{z}[x, y]$ and $\mathbf{N}_y[x, y] = \mathbf{z}[x, y+1] - \mathbf{z}[x, y]$. I don't explicitly write it out.